

Load combinations which include settlement shall also be applied without settlement. As specified in Article 3.12.6, differential movements between and within substructure units shall be considered when determining the most critical combinations of force effects.

Downdrag shall be evaluated at the service limit state using the provisions of Article 3.11.8.

For segmentally constructed bridges, the following combination shall be investigated at the service limit state:

$$DC + DW + EH + EV + ES + WA + CR + SH + TG + EL + PS$$

(3.4.1-2)

Table 3.4.1-1—Load Combinations and Load Factors

Load Combination Limit State	DC DW EH EV ES EL PS CR SH	LL IM CE BR PL LS	WA	WS	WL	FR	TU	TG	SE	DR	Use One of These at a Time				
											EQ	BL	IC	CT	CV
Strength I (unless noted)	γ_p	1.75	1.00	—	—	1.00	γ_{TU}	γ_{TG}	γ_{SE}	γ_{DR}	—	—	—	—	—
Strength II	γ_p	1.35	1.00	—	—	1.00	γ_{TU}	γ_{TG}	γ_{SE}	γ_{DR}	—	—	—	—	—
Strength III	γ_p	—	1.00	1.00	—	1.00	γ_{TU}	γ_{TG}	γ_{SE}	γ_{DR}	—	—	—	—	—
Strength IV	γ_p	—	1.00	—	—	1.00	γ_{TU}	—	—	—	—	—	—	—	—
Strength V	γ_p	1.35	1.00	1.00	1.00	1.00	γ_{TU}	γ_{TG}	γ_{SE}	γ_{DR}	—	—	—	—	—
Extreme Event I	1.00	γ_{EQ}	1.00	—	—	1.00	—	—	—	1.00	1.00	—	—	—	—
Extreme Event II	1.00	0.5/ 1.00	1.00	—	—	1.00	—	—	—	1.00	—	1.00	1.00	1.00/ 1.00	1.00
Service I	1.00	1.00	1.00	1.00	1.00	1.00	γ_{TU}	γ_{TG}	γ_{SE}	1.00	—	—	—	—	—
Service II	1.00	1.30	1.00	—	—	1.00	γ_{TU}	—	—	—	—	—	—	—	—
Service III	1.00	γ_{LL}	1.00	—	—	1.00	γ_{TU}	γ_{TG}	γ_{SE}	1.00	—	—	—	—	—
Service IV	1.00	—	1.00	1.00	—	1.00	γ_{TU}	—	1.00	1.00	—	—	—	—	—
Fatigue I—LL, IM & CE only	—	1.75	—	—	—	—	—	—	—	—	—	—	—	—	—
Fatigue II—LL, IM & CE only	—	0.80	—	—	—	—	—	—	—	—	—	—	—	—	—

Notes: For Service I, the load factor for *EV* equals 1.2 for Stiffness Method Soil Failure as shown in Table 3.4.1-2.
The value of γ_p shall be taken from Tables 3.4.1-2 and 3.4.1-3.
The value of γ_{LL} shall be taken from Table 3.4.1-4.
The value of γ_{SE} shall be taken from Table 3.4.1-5.
The value of γ_{TU} shall be taken from Table 3.4.1-6.
For Structural Strength, the load factor for *DR*, γ_{DR} , equals 1.10 and for Geotechnical Strength, γ_{DR} equals 0.00.